# Demo: Forecasting a Café's Daily Ingredient Demand With Damped Triple Exponential Smoothing

**Bastian Queckenstedt · Tobias Christopher Kröncke · Efe Enes Nayci**
Department of Informatics, Universität Hamburg

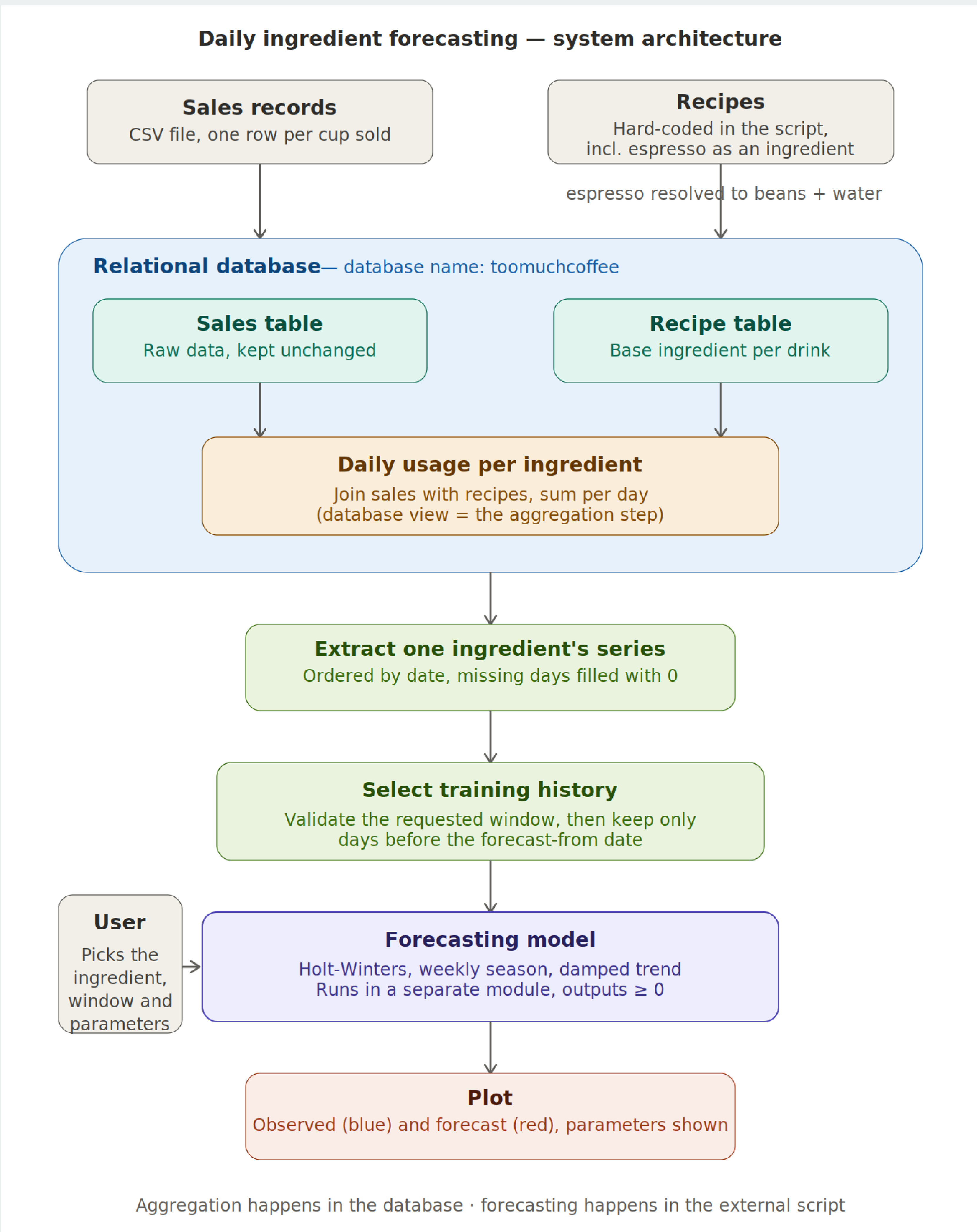
## The problem

- A café has records on how many cappuccinos, cocoas, espressos etc. it sold in the past.
- It must purchase the right amounts of what it will consume in the future: **Coffee beans, milk, water, cocoa powder, sugar, salt.**

## What the demo does

- Lightweight Forecasting Pipeline
- One year of real sales (3 547 cups, Mar 2024 – Mar 2025) is loaded **unchanged** into PostgreSQL.
- Recipes are resolved into **six base ingredients** (espresso shots → 18 g beans + 30 ml water; "milk or water" → water).
- A SQL **view** joins sales with recipes and aggregates the daily usage per ingredient – declaratively, inside the database.
- A Python CLI extracts one series, forecasts it, and plots the result automatically.

## Architecture




## How: damped Holt–Winters

Additive triple exponential smoothing, implemented from scratch, season length $m = 7$ days:

$$\ell_t = \alpha(x_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + \phi b_{t-1})$$
$$b_t = \beta(\ell_t - \ell_{t-1}) + (1-\beta)\,\phi b_{t-1}$$
$$s_t = \gamma(x_t - \ell_t) + (1-\gamma)\,s_{t-m}$$

- Damping $\phi < 1$ lets the trend level off instead of exploding over long horizons.
- Forecasts are clipped at zero – negative usage is impossible.

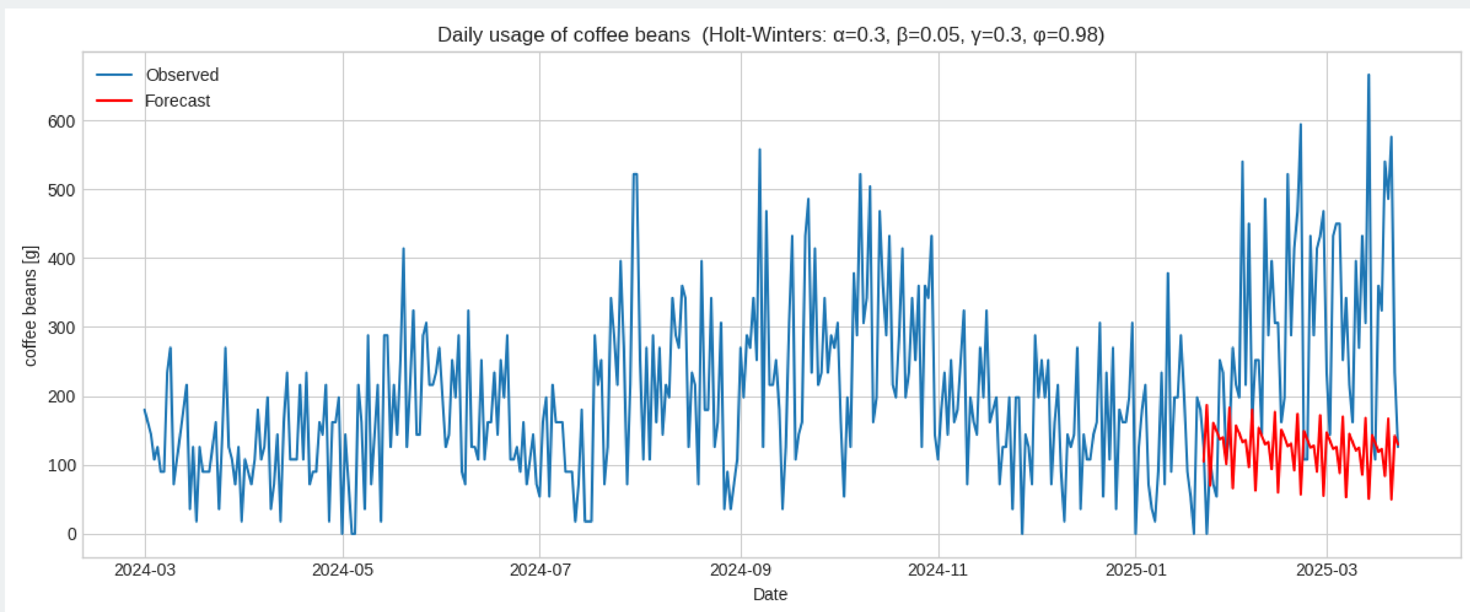
## Try it yourself!

Switch the ingredient, move the window, explore undamped limit:

```
$ ./timeseries.py -i "coffee beans" -a 0.3 -b 0.05 -g 0.3 -p 0.98
$ ./timeseries.py -i salt
$ ./timeseries.py -i milk -fu "2026-07-14"
$ ./timeseries.py -i milk -fu "2026-07-14" -p 1.0     (undamped trend!)
```

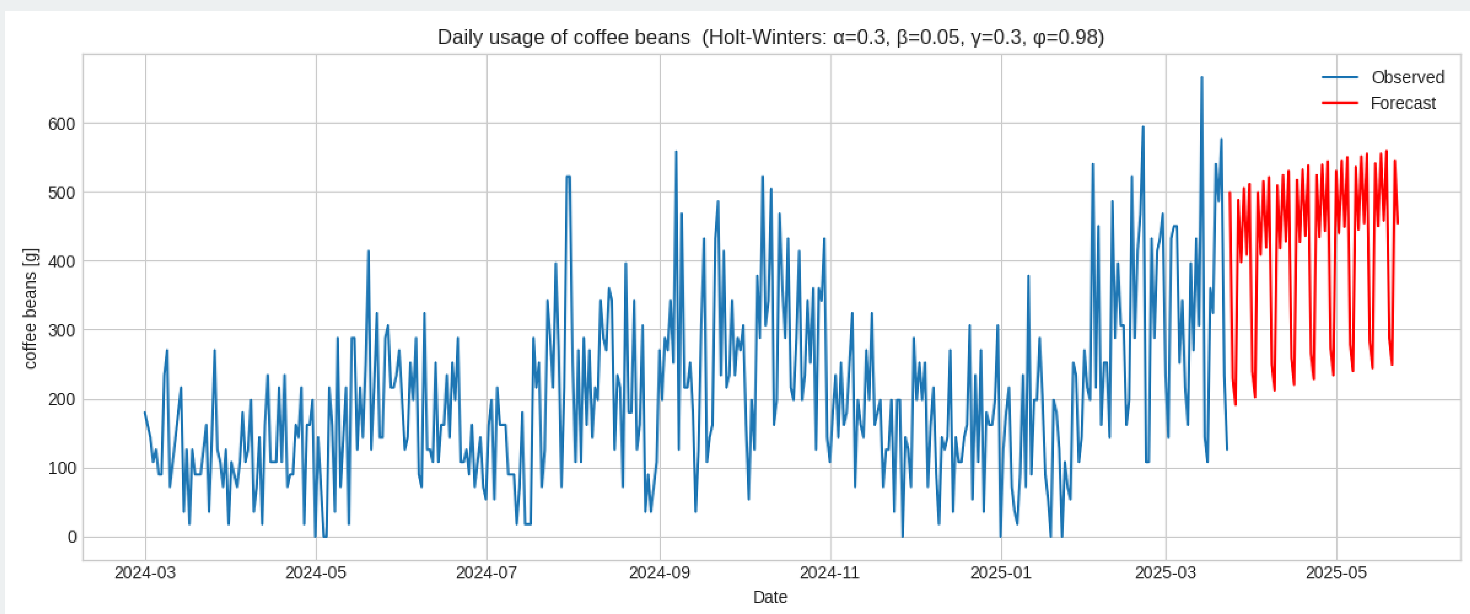
- All four smoothing parameters $(\alpha, \beta, \gamma, \phi)$ are command line flags.
- Forecasting window defined by `-ff` ("forecast from") and `-fu` ("forecast until")

## Does it work? Validation




The model forecasts the **last two observed months** using only the history before the window.

## Looking ahead




Two-month forecast **beyond the data**: the answer to the café's actual planning question – how many beans to buy next week.

Method: Holt (1957/2004), Winters (1960), damped trend: Gardner & McKenzie (1985) · Data: course datasets, DIS SuSe 2026, Universität Hamburg · Stack: PostgreSQL, Python (psycopg2, matplotlib), Tuxedo OS